Figure 7: The IoT platform of Carriots

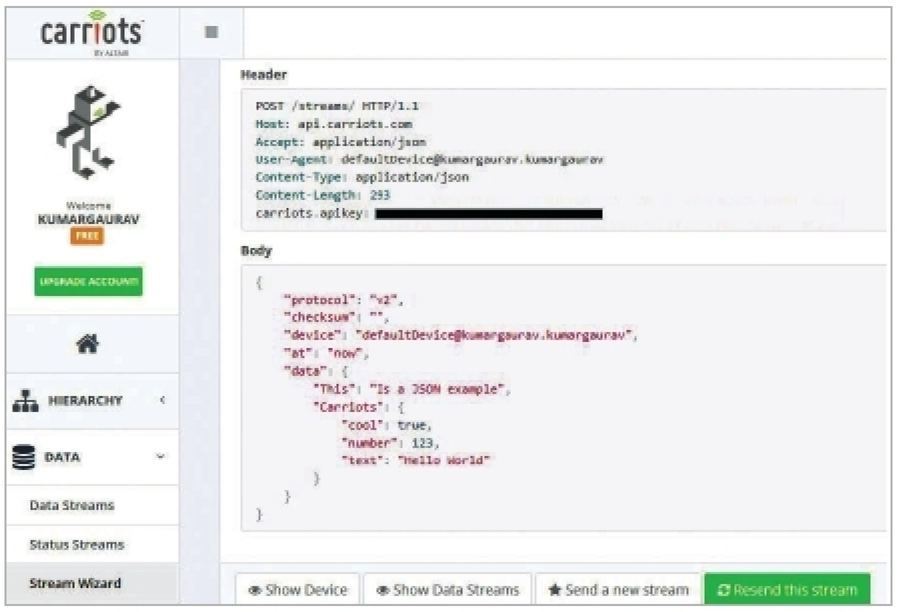

Figure 8: Generating an API key from the dashboard of Carriots

panel is provided that has options for creating and mapping devices and streams. There are options to integrate the streaming with e-mail and SMS for live notifications.

The following is the source code for the Arduino IDE which transmits data to Carriots:

```
#include “ESP8266WiFi.h”
const char* WiFiNetworkName     = “<WiFi Network Name>”;
const char* NetworkPassword = “<WiFi NetworkPassword>”;
const char* ServerName = “api.carriots.com”;
const String CARRIOTSAPI = “<API Key of Carriots>”;
const String DEVICE = “defaultDevice@kumargaurav.
kumargaurav”;
WiFiClient client;
int readval = 0;
void setup() {
  Serial.begin(9600);
  delay(1000);
  Serial.println();
  Serial.print(“Connecting to Network”);
  Serial.println(WiFiNetworkName);
  WiFi.begin(WiFiNetworkName, NetworkPassword);
  while (WiFi.status()!= WL_CONNECTED) {
    delay(1000);
  }
  Serial.println(“Connected”);
}
void sendStream()
{
  if (client.connect(ServerName, 80)) {
    Serial.println(F(“connected”));
    String json = “{\”protocol\”:\”v2\”,\”device\”:\”” +
DEVICE + “\”,\”at\”:\”now\”,\”data\”:{\”RecordedData\”:\”” +
val + “\”}}”;
    client.println(“POST /streams HTTP/1.1”);
    client.println(“Host: api.carriots.com”);
    client.println(“Accept: application/json”);
    client.println(“User-Agent: Arduino-Carriots”);
    client.println(“Content-Type: application/json”);
    client.print(“carriots.CarriotsAPI: “);
    client.println(CARRIOTSAPI);
    client.print(“Content-Length: “);
    int thisLength = json.length();
    client.println(thisLength);
    client.println(“Connection: close”);
    client.println();
    client.println(json);
  }
  else {     Serial.println(F(“Not Connected”));  }
}
void loop() {
  readval = analogRead(A0);
  Serial.println(val);
  Serial.println(F(“Transmit Data”));
  sendStream();
  delay(1000);
}
```

After executing the code in the Arduino IDE, the real-time data is transmitted to the Carriots server for recording. There are multiple file formats in which data can be stored for further evaluation.

The tech world is trying to connect every ‘thing’ using network based devices. This type of communication in which all objects are connected with each other is referred to as the Internet of Things (IoT). To work in this domain, there is a need for real sensor based devices with customisable hardware and chips. By using these, a well connected and programmed environment can be created in which secure and effective communication can take place. There is a lot of open source hardware available, including Arduino and Raspberry Pi, which can be programmed for multiple real-life applications for low cost IoT implementations. END

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